

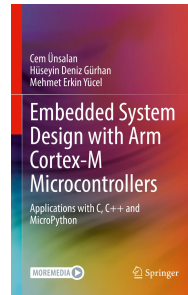
ELK 346 E Data Acquisition & Embedded Systems

2023 – 2024 Spring Semester
Dr. Aydın Tarık Zengin
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Resources

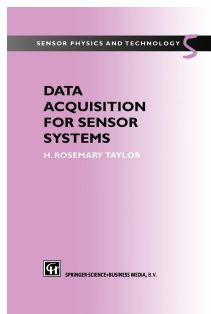


- Cem Ünsalan, Hüseyin Deniz Gürhan, Mehmet Erkin Yücel; Embedded System Design with Arm Cortex-M Microcontrollers, 2022, Springer

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Resources

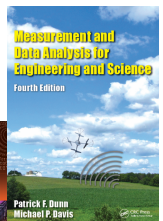
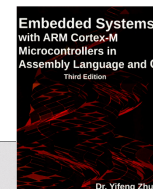
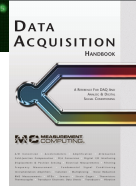


- H. Rosemary Taylor; Data Acquisition for Sensor Systems, 1997, Springer

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Other Resources



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Topics

- Sensors
- Filters, ADCs, DACs
- Sampling
- Fundamental Digital Electronics
- Basics of Microcontrollers – Registers, ALU
- ARM - Introduction
- ARM – Registers
- Embedded Programming

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Evaluation

Midterm Exam	%30
Homework	%10
Term Project	%20
Final	%40

- The Midterm will be held on April 5th.
- One or more homework will be given throughout the semester
- You must submit and present a term project.
- To attend the final exam, you **MUST** submit and present the **working** term project and homework on time.

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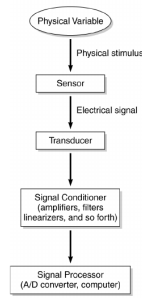
Term Project Details

- Form a project group of **3-4 students** by the **3rd week** of the semester. (Groups with less than 3-students will not be allowed due to the high number of students taking the class)
- Group members will be registered and **won't be changed!**
- Group will determine a term project idea by the **6th week**
- Projects will be completed by the 10th week and a project report must be submitted. (Reports will be written in LaTeX and the details will be announced later)**
- Project groups should obtain the materials such as embedded development board and sensors by themselves.
- All the term projects will be **presented by the project groups in weeks 11-14**
- The first and most important evaluation criteria is a **working project**.

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Sensors



Temperature, pressure, strain, length, and so forth.

Temperature

Temperature >> resistance:
 $R = R_0 (1 + \alpha(T - T_0))$

Resistance

Resistance >> voltage
(using Wheatstone Bridge)

Signal typically a voltage

Signal filtered and gained to proper frequency, range and magnitude.

Analog signal sampled and converted to digital signal with proper resolution. Digital signal stored and possibly analyzed.

- Sensor** is a device that senses a physical stimulus and converts it into a signal
- The device that changes (transduces) the signal into the desired quantity (be it electrical, mechanical, optical, or another form) is the **transducer**. In the most general sense, a transducer transforms energy from one form to another.

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Sensor Domains

- Sensors can be categorized into domains by **what they sense**. These domains and the sensed variables include;
- chemical**: chemical concentration, composition, and reaction rate,
- electrical**: current, voltage, resistance, capacitance, inductance, and charge,
- magnetic**: magnetic field intensity, flux density, and magnetization,
- mechanical**: displacement or strain, level, position, velocity, acceleration, force, torque, pressure, and flow rate,
- radiant**: electromagnetic wave intensity, wavelength, polarization, and phase, and
- thermal**: temperature, heat, and heat flux.

Try to find some actual sensors in the categories above, download datasheets.

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Sensor Domains

- Sensors also can be organized according to the physical mode of **how they sense**. These modes are
- electric**,
- piezoresistive**,
- fluid mechanic**,
- optic**,
- photoelastic**,
- thermoelectric**,
- electrochemical**.

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Electric

- Sensors in this category are based upon a **change in either resistance, capacitance, or inductance** that results from applying a physical stimulus to the sensor.
- Their output, in some instances, can be read directly using an ohmmeter, or the voltage drop on it in a voltage-divider circuit.

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Resistive

- A sensor based upon the principle that a change in resistance can be produced by a change in a physical variable
- A resistance sensor can be used to measure **displacement, strain, force, pressure, acceleration, flow velocity, temperature, heat or light flux, gas concentration, etc.**

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Resistive

- **Resistance Temperature Detector - RTD**
- α_T is coefficient of thermal expansion, and R_o is the resistance at the reference temperature T_o .

$$T = T_o + \left[\frac{R - R_o}{\alpha_T R_o} \right]$$

- The metals usually used are platinum, copper, or nickel, having coefficients of thermal expansion, α_T , of $0.0039/^\circ\text{C}$, $0.0038/^\circ\text{C}$, or $0.0067/^\circ\text{C}$, respectively.

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Resistive

- Example: A **platinum** RTD is used to determine the liquid temperature of a yeast-proofing mixture. The measured resistance of the RTD is 25.0Ω at 0°C and 29.8Ω at the proofing temperature. What is the proofing temperature (in K)?

- Solution:

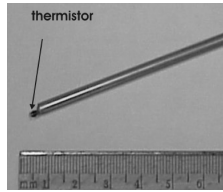
$$T(^{\circ}\text{C}) = 0 + (29.8 - 25.0) / [(0.0039)(25.0)] = 49.2 ^{\circ}\text{C} = 322.4 \text{ K}$$

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Resistive: Thermistor

- A thermistor is another sensor that is used to measure temperature. It consists of two wires that are connected across a small piece of semiconductor material
- Thermistors can have either a decreasing or an increasing resistance with temperature. The former is called a **negative temperature coefficient (NTC)** thermistor and the latter a **positive temperature coefficient (PTC)** thermistor.

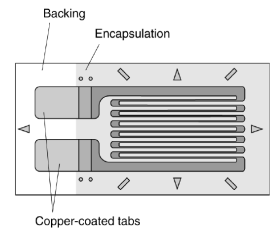


$$T = \left[\frac{T_o}{(T_o/\beta_T) \log_e(R/R_o) + 1} \right]$$

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Resistive: Strain Gauge

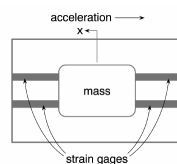
- A Strain gauge is a sensor whose **resistance varies with applied force**.
- It converts force, pressure, tension, weight, etc., into a change in electrical resistance which can then be measured.



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Resistive: Resistive Accelerometer

- A resistive accelerometer uses a strain gage flexure arrangement
- The system essentially behaves as a spring-mass system.



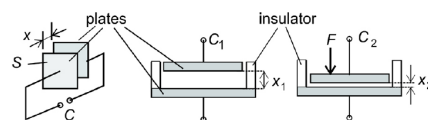
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Capacitive

- Capacitive sensors produce a **change in capacitance** with the change in a physical variable.
- A capacitive sensor consists of **two small conducting plates**, each of area, A_{cp} , separated by a distance, h , with a dielectric material in between the two plates. The capacitance between the two plates is

$$C = \epsilon_o \epsilon A_{cp} / h$$

- in which ϵ_o is the permittivity of free space ($= 8.85 \times 10^{-12} \text{ F/m}$) and ϵ the relative permittivity.



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Capacitive Sensors

- Capacitive (Variable Spacing) Displacement Indicator
- Capacitive (Variable Area) Displacement Indicator
- Capacitive (Variable Dielectric) Displacement Indicator
- Capacitive Pressure Transducer
- Capacitive Accelerometer
- Capacitor Microphone
- Relative Humidity Sensor

Some
Research
Required!

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Piezoelectric

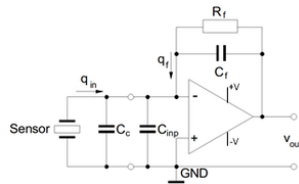
- Piezoelectric materials, like **quartz** and **polyvinyl difluoride (PVDF)**, develop an electric charge and, hence, an electric field when deformed. This is called the **direct piezoelectric effect**.
- Conversely, when an electric field or voltage is applied to the material, it deforms. This is called the **converse (or reverse) piezoelectric effect**.
- When a **mechanical stress** is applied to a piezoelectric material a **strain occurs**, thereby producing a displacement of ions within the stressed material.

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Charge Amplifier

- The **charge amplifier** is an operational amplifier whose positive and negative inputs are the outputs of the two electrodes of a **piezoelectric sensor**.
- It's a **current integrator** that produces a voltage output proportional to the integrated value of the input current, or the **total charge injected**.

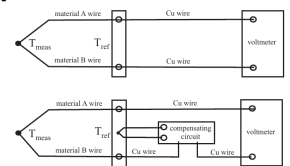


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Thermocouple

- When segments of wires consisting of **two dissimilar metals** are joined at each of their two ends and the ends are **exposed to different temperatures**, an **emf** will be generated in an open circuit formed by having a break in one of the wires.
- The emf varies with the type of metal and the temperature difference.

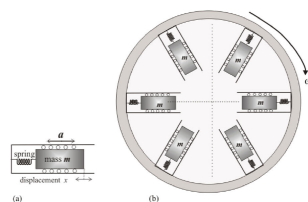


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Gyroscope

- Gyroscopes are used for detection and measurement of **orientation of objects in rotational motion or with changing orientation (Angular velocity)**
- Gyroscopes in consumer electronics are frequently **combined with accelerometers** for more robust direction- and motion-sensing.



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Research Topic

- Describe the actual **sensor** in each of the following systems and the basic physical principle or law upon which each works:
 - (a) a tire pressure gauge,
 - (b) a toaster,
 - (c) a car mph/kph indicator,
 - (d) a television remote control, and
 - (e) a battery-life indicator on a digital camera.

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